ELEC2134: Class Test 4 (System Analysis)

Date: Thursday 19 March 2026 (Week 5)

Time: 3 - 4 pm (Exam Duration 40 minutes).

Instruction to candidates: To maximise your marks, show all steps clearly when deriving your answers (do not write final answers only).

Pass Level Questions. Total Marks: 64

Question 1:

A system provides an ideal delay of T seconds, and the input to the system $x(t)$ and output $y(t)$ are related by $y(t) = x(t - T)$.

(i) Find the transfer function of the system **[10 Marks]**

(ii) Sketch the magnitude and Phase responses of the system. **[20 Marks]**

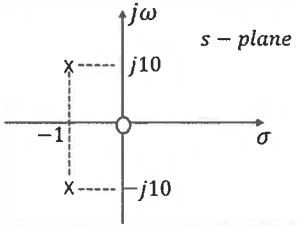
Question 2:

The pole–zero plot of a system is shown below. The system has a pair of complex conjugate poles and a zero at the origin.

(a) Determine the transfer function from the pole–zero plot. **[15 Marks]**

(b) Find the DC value of the system and its magnitude at $\omega = 10$ rad/s. **[10 Marks]**

(c) Sketch the magnitude response of the system, showing key values clearly. **[14 Marks]**



Note: You may state any assumption(s) made.

Higher Level Questions. Total Marks: 36

Question 3:

A transfer function of a lowpass filter is given by $H(s) = \frac{\omega_p^2}{s^2 + b_p s + \omega_p^2}$. Show that the magnitude response at $\omega = \omega_p$ is as given by $|H(j\omega)| = Q$, where $Q(\text{Quality factor}) = \frac{\omega_p}{b_p}$.

[16 Marks]

Question 4:

The transfer function of a system is given by $H(s) = \frac{\omega_c}{s + 10\omega_c}$. Show that the 3-db cut-off frequency of the system is $10\omega_c$.

[20 Marks]

Laplace Transform Formula Sheet for Week 5 Exam

Basic Laplace Transforms

Signal	Transform
$u(t)$	$\frac{1}{s}$
$t\,u(t)$	$\frac{1}{s^2}$
$t^n u(t)$	$\frac{n!}{s^{n+1}}$
$\frac{t^{n-1}}{(n-1)!}u(t)$	$\frac{1}{s^n}$
$\delta(t-\tau),\ \tau>0$	$e^{-s\tau}$
$e^{-at}u(t)$	$\frac{1}{s+a}$
$te^{-at}u(t)$	$\frac{1}{(s+a)^2}$
$\cos(\omega t)u(t)$	$\frac{s}{s^2+\omega^2}$
$\sin(\omega t)u(t)$	$\frac{\omega}{s^2+\omega^2}$
$e^{-at}\cos(\omega t)u(t)$	$\frac{s+a}{(s+a)^2+\omega^2}$
$e^{-at}\sin(\omega t)u(t)$	$\frac{\omega}{(s+a)^2+\omega^2}$

Laplace Transform Properties

Signal	Transform
$x(t)$	$X(s)$
$ax(t)+by(t)$	$aX(s)+bY(s)$
$x(t-\tau)$	$e^{-s\tau}X(s)$
$e^{a_0 t}x(t)$	$X(s-a_0)$
$x(at),\ a>0$	$\frac{1}{a}X\left(\frac{s}{a}\right)$
$x(t)*y(t)$	$X(s)Y(s)$
$-tx(t)$	$\frac{d}{ds}X(s)$
$\frac{dx(t)}{dt}$	$sX(s)-x(0)$
$\frac{d^2x(t)}{dt^2}$	$s^2X(s)-sx(0)-x'(0)$

Class Test 4

Model Answers

Question 1

$$y(t) = x(t-T)$$

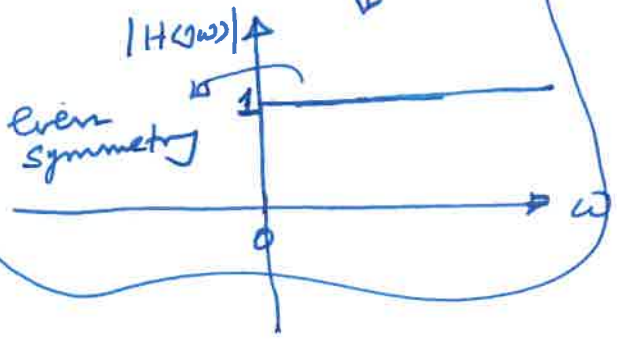
$$Y(s) = X(s)\, e^{-sT}$$

$$\therefore H(s) = \frac{Y(s)}{X(s)} = e^{-sT}$$ — 10 Marks

$$H(j\omega) = H(s)\Big|_{s=j\omega} = e^{-j\omega T}$$

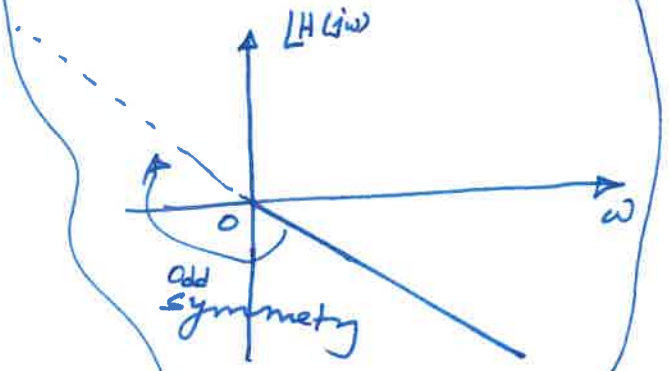
$$\therefore |H(j\omega)| = 1$$

10 Marks



10 Marks

$$\& \quad \angle H(j\omega) = -\omega T$$



Q2

(a) 15 Marks

$$H(s) = \frac{s}{[s-(-1+j10)][s-(-1-j10)]} = \frac{s}{(s+1)^2+10^2}$$

$$H(s) = \frac{s}{s^2+2s+101}$$

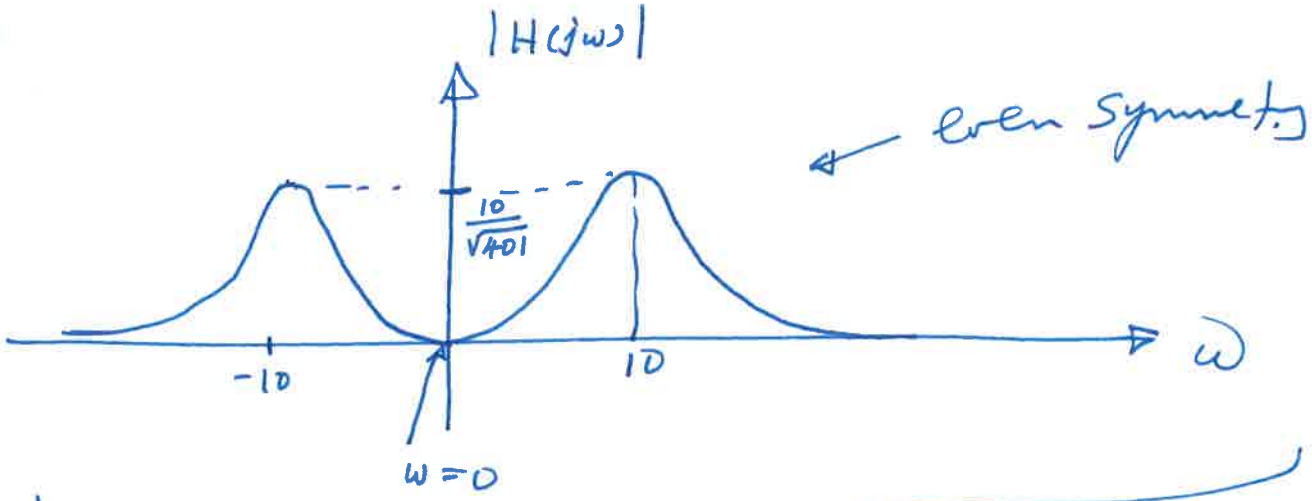
(b)

$$|H(j\omega)| = \frac{|j\omega|}{|(j\omega+1)^2+10^2|} = \frac{\omega}{|j\omega+1-j10|\,|j\omega+1+j10|}$$

$$|H(0)| = |H(j\omega)|_{\omega=0} = 0$$ — DC component — 5 Marks

$$|H(j\omega)|_{\omega=10} = \frac{10}{(j10+1-j10)(j10+1+j10)} = \frac{10}{\sqrt{401}}$$ — 5 Marks

(c)



14 Marks

Question 3

$$H(s) = \frac{\omega_p^2}{s^2 + b_p s + \omega_p^2} \qquad b_p = \frac{\omega_p}{Q}$$

16 Marks

$$H(j\omega) = \frac{\omega_p^2}{(j\omega)^2 + \frac{\omega_p}{Q} j\omega + \omega_p^2}$$

$$|H(j\omega)| = \frac{\omega_p^2}{\sqrt{(\omega_p^2 - \omega^2)^2 + \left(\frac{\omega\omega_p}{Q}\right)^2}}$$

When $\omega = \omega_p$

$$|H(j\omega_p)| = \frac{\omega_p^2}{\sqrt{0 + \frac{\omega_p^4}{Q^2}}} = \frac{\omega_p^2 \times Q}{\omega_p^2} = Q$$

Question 4

20 Marks

$$H(s) = \frac{\omega_c}{s + 10\omega_c} \quad \therefore |H(j\omega)| = \left|\frac{\omega_c}{j\omega + 10\omega_c}\right|$$

$$|H(j\omega)| = \frac{\omega_c}{\sqrt{\omega^2 + 100\omega_c^2}}; \quad \text{when } \omega = 0, |H(j0)| = \frac{\omega_c}{10\omega_c} = \frac{1}{10} \checkmark$$

For 3 dB cut off, $|H(j\omega)|_\omega = \frac{1}{\sqrt{2}} |H(j0)| \Rightarrow \frac{1}{\sqrt{2}} \cdot \frac{1}{10} = \frac{\omega_c}{\sqrt{\omega^2 + 100\omega_c^2}}$

$$200\omega_c^2 = \omega^2 + 100\omega_c^2 \Rightarrow \boxed{\omega = 10\omega_c}$$